1. Create a Flask application with an `/api` route. When this route is accessed, it should return a JSON list. The data should be stored in a backend file, read from it, and sent as a response.

⇒

1) Create JSON file data.json

```
NEW FOLDER
  python_tutorial
    assignment1
    assignment2
      app.py
      data.json
```

python_tutorial > assignment2 > {} data.json > ...

```
[
    {"id": 1, "name": "Harsha", "role": "DevOps Engine
    {"id": 2, "name": "Divya", "role": "Team Lead"},
    {"id": 3, "name": "Amit", "role": "Developer"}
]
```

2) create Flask App file app.py

python_tutorial > assignment2 > app.py > ...

```python
from flask import Flask, jsonify
import json

app = Flask(__name__)

# Home route
@app.route('/')
def home():
    return "Flask API is running! Use /api to get data"

# API route
@app.route('/api', methods=['GET'])
def get_data():
    try:
        # Read data from JSON file
        with open('data.json', 'r') as file:
            data = json.load(file)

        return jsonify(data)

    except Exception as e:
        return jsonify({"error": str(e)}), 500

# Run the app
if __name__ == '__main__':
    app.run(debug=True)
```

3) You can see the output like:

* Running on http://127.0.0.1:5000

4) Open Browser and go to:

http://127.0.0.1:5000/api

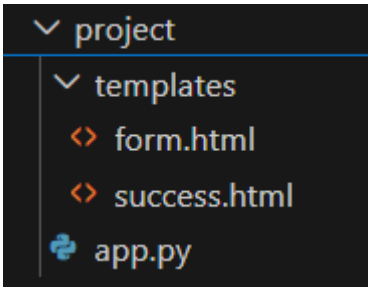
5) You will see the JSON data output.

2. Create a form on the frontend that, when submitted, inserts data into MongoDB Atlas. Upon successful submission, the user should be redirected to another page displaying the message "Data submitted successfully". If there's an error during submission, display the error on the same page without redirection.

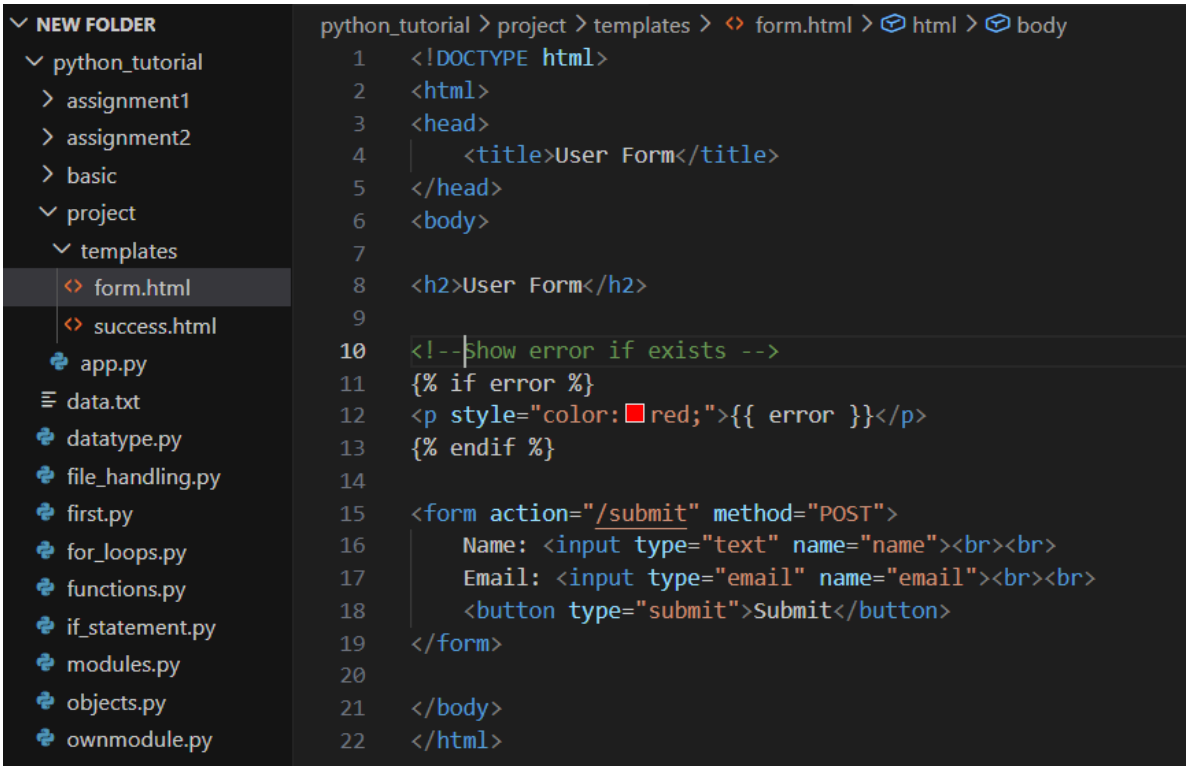
⇒

First for the above project, need to create cluster

1) Go to https://www.mongodb.com/atlas
2) Create account
3) Add new database user
4) Go to database section
5) Choose Driver - Python
6) One connection string is showing there copy it.
7) For creating this project, My folder structure is like below:



Form.html:



Success.html:

```
EXPLORER
NEW FOLDER
  python_tutorial
    assignment1
    assignment2
    basic
    project
      templates
        form.html
        success.html
      app.py
    data.txt
```

python_tutorial > project > templates > success.html > html

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Success</title>
5  </head>
6  <body>
7
8  <h2>Data submitted successfully</h2>
9
10 </body>
11 </html>
```

app.py:

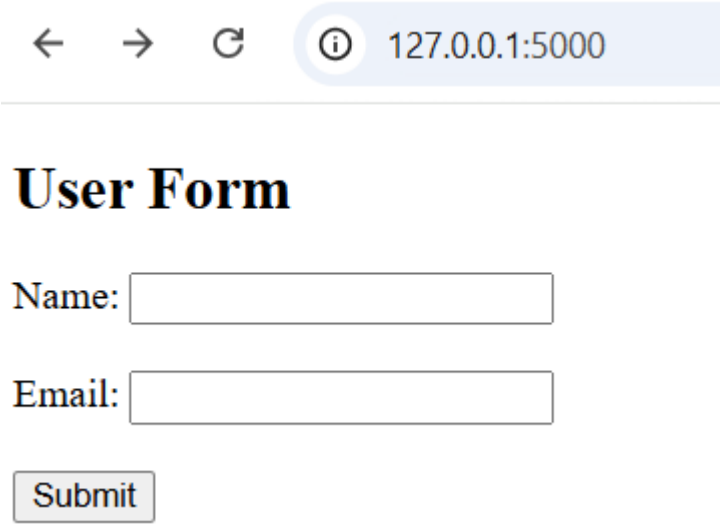
python_tutorial > project > app.py > ...

```
1  from flask import Flask, render_template, request, redirect, url_for
2  from pymongo import MongoClient
3
4  app = Flask(__name__)
5
6  # MongoDB Atlas connection (my string)
7  client = MongoClient("mongodb+srv://awsharshu2209_db_user:QIUIrYWdDI9wc3ub@cluster0.e7rttym.mongodb.net/?appName=Cluster0")
8
9  # Database & Collection
10 db = client["mydatabase"]
11 collection = db["users"]
12
13 # Form page
14 @app.route('/')
15 def form():
16     return render_template('form.html')
17
18 # Handle form submission
19 @app.route('/submit', methods=['POST'])
20 def submit():
21     try:
22         name = request.form['name']
23         email = request.form['email']
24
25         # Validation
26         if not name or not email:
27             raise Exception("All fields are required!")
```

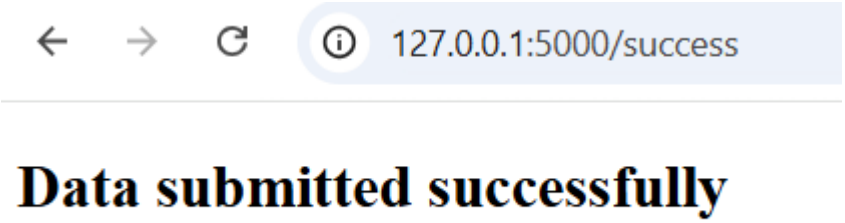
8) Here, in line no. 7 copy that connection string that you copied earlier.

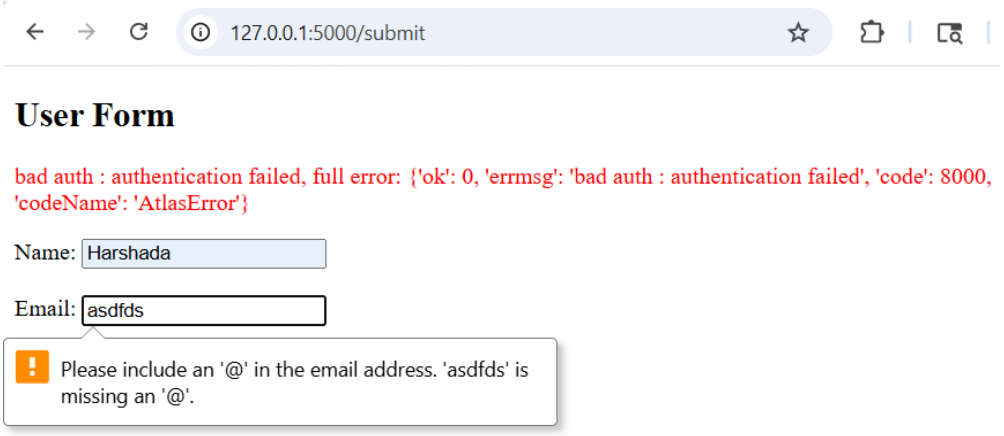
```python

        # Insert into MongoDB
        collection.insert_one({
            "name": name,
            "email": email
        })

        # Success → redirect
        return redirect(url_for('success'))

    except Exception as e:
        # Error → stay on same page
        return render_template('form.html', error=str(e))

# Success page
@app.route('/success')
def success():
    return render_template('success.html')

# Run app
if __name__ == '__main__':
    app.run(debug=True)
```

9) Now, Run the application (Python app.py)
10) On the terminal we can see the URL: * Running on http://127.0.0.1:5000
11) Open that URL on browser
12) You can see our output i.e. user form



13) After filling that User form with correct name and Email, the below message is showing:



Data submitted successfully

14) If we insert the wrong data, the below msg is showing on the same window without redirecting to another window